Quantification: 699_HYP_HP_z20c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	20
interior lacuna count	16
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 16)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1865.56	1420	1569.24	px^2
major_axis_length_px	lacuna shape	84.1782	74.9186	43.6226	px
minor_axis_length_px	lacuna shape	34.5802	31.386	17.224	px
aspect_ratio	lacuna shape	2.6998	2.5248	1.2089	unitless
eccentricity	lacuna shape	0.8681	0.9173	0.1156	unitless
solidity	lacuna shape	0.7566	0.7699	0.1071	unitless
roots_count	headline	6.4375	5.5	3.7589	count
ring_length_r30_px	headline	222.812	205	148.529	px
ring_length_r60_px	reported	450.438	413	319.068	px
owned_length_px	ownership-dependent	485.533	265.388	493.269	px
edge_count	ownership-dependent	26.125	16.5	25.411	count
mean_edge_length_px	ownership-dependent	18.1734	19.0637	5.2023	px
perimeter_px	normalised	259.394	222.853	140.776	px
ring_area_r30_px2	normalised	8103.69	7406.5	3385.63	px^2
ring_area_r60_px2	normalised	18401.4	18083	6926.79	px^2
in_frame_fraction_r30	normalised	0.9852	1	0.0593	unitless
in_frame_fraction_r60	normalised	0.9791	1	0.0834	unitless
ring_density_r30	normalised	0.0284536	0.0295362	0.0176441	px^-1
ring_density_r60	normalised	0.0239362	0.0275446	0.0144993	px^-1
roots_per_100px_perimeter	normalised	2.9499	2.3064	1.9137	per 100 px
ring_attached_length_r30_px	network v2	181.312	161.5	138.144	px
ring_attached_length_r60_px	network v2	301.312	217.5	248.456	px
ring_length_w_r30_px	network v2	248.095	222.675	168.984	px
ring_length_w_r60_px	network v2	504.074	458.469	361.087	px
sholl_crossings_r10	network v2	5	5	3.7059	count
sholl_crossings_r20	network v2	5.8125	5	4.1508	count
sholl_crossings_r30	network v2	6.75	6	4.3282	count
loop_count	loops, report only	1	0	2.3094	count
loop_edge_share	loops, report only	0.0727	0	0.1357	share
ring_width_mean_r30_px	width	6.8259	6.8215	0.9156	px

Per field
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measure	value	units
analysed_area_px2	993977	px^2
total_skeleton_length_px	17917	px
canalicular_length_density_per_px	0.0180256	px^-1
junction_count	355	count
junction_density_per_px2	0.00035715	px^-2
lacuna_count	20	count

lacunae_per_px2	2.012e-05	px^-2
skeleton_component_count	323	count
mean_component_length_px	55.4706	px
median_component_length_px	19	px
field_length_density_w_per_px	0.0201822	px^-1
field_density_without_flagged_per_px	0.0169761	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	1026	px
unattached_component_count	51	count
canalicular_length_density_attached_per_px	0.0180256	px^-1
field_length_density_with_unattached_per_px	0.0190578	px^-1
loop_count	31	count
loop_edge_share	0.1227	share
edge_count	1019	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.997	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	13630	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	582	37.2403	23.3143	1.5973	0.7798	0.8362	-1.3908	9.6323	718.978	True
2	701	53.6823	22.5905	2.3763	0.9071	0.7242	1.3601	138.937	341.827	False
3	1079	46.6189	34.6236	1.3464	0.6696	0.7599	-0.6851	205.172	387.171	False
4	3222	86.3749	58.8691	1.4672	0.7318	0.7263	-0.9453	298.969	636.32	False
5	4692	101.01	74.3902	1.3578	0.6765	0.6687	0.5065	385.336	36.169	True
6	12231	202.474	100.48	2.0151	0.8682	0.7797	0.0478	526.667	38.6221	True
7	1329	56.9074	38.8202	1.4659	0.7312	0.7408	-0.6935	386.281	385.088	False
8	574	63.7	12.3761	5.147	0.9809	0.8658	-1.4148	410.084	638.57	False
9	961	72.5694	18.5814	3.9055	0.9667	0.8076	1.4502	539.664	699.95	False
10	1511	109.373	23.0638	4.7422	0.9775	0.6218	1.4725	559.445	447.167	False
11	599	49.8148	18.0062	2.7665	0.9324	0.7799	1.3874	565.563	692.351	False
12	898	39.9158	30.7419	1.2984	0.6378	0.8635	1.334	631.296	609.345	False
13	2133	94.4671	56.7385	1.665	0.7995	0.538	0.9278	716.599	755.109	False
14	2065	77.2677	36.7006	2.1054	0.88	0.8765	-0.6191	783.319	730.515	False
15	2136	110.111	32.0301	3.4377	0.9568	0.7495	-0.4861	842.146	688.921	False
16	7245	127.692	88.342	1.4454	0.7221	0.8447	-1.2489	821.694	971.904	True
17	7049	221.75	78.7671	2.8153	0.9348	0.5533	-1.5418	863.529	297.474	False
18	2325	112.289	29.7925	3.769	0.9642	0.7992	1.5515	897.959	158.373	False
19	2028	91.0023	34.0428	2.6732	0.9274	0.8145	0.9697	910.357	523.35	False
20	1239	61.0069	27.5385	2.2153	0.8923	0.8844	-1.1926	1001.21	883.603	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

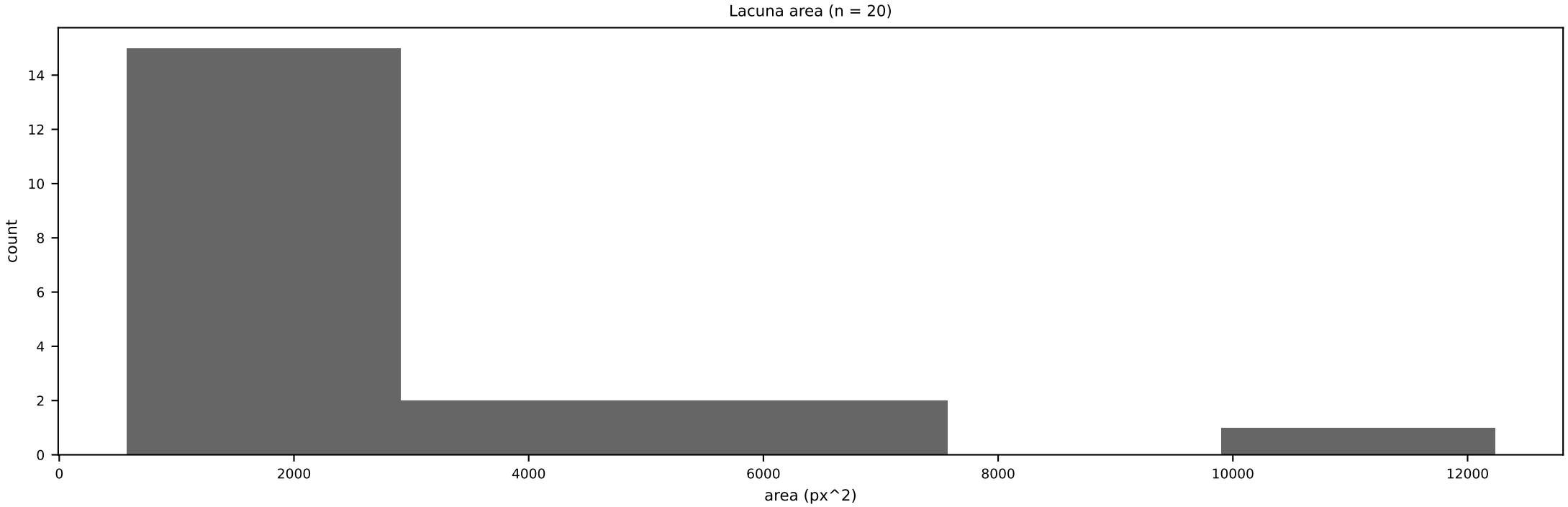
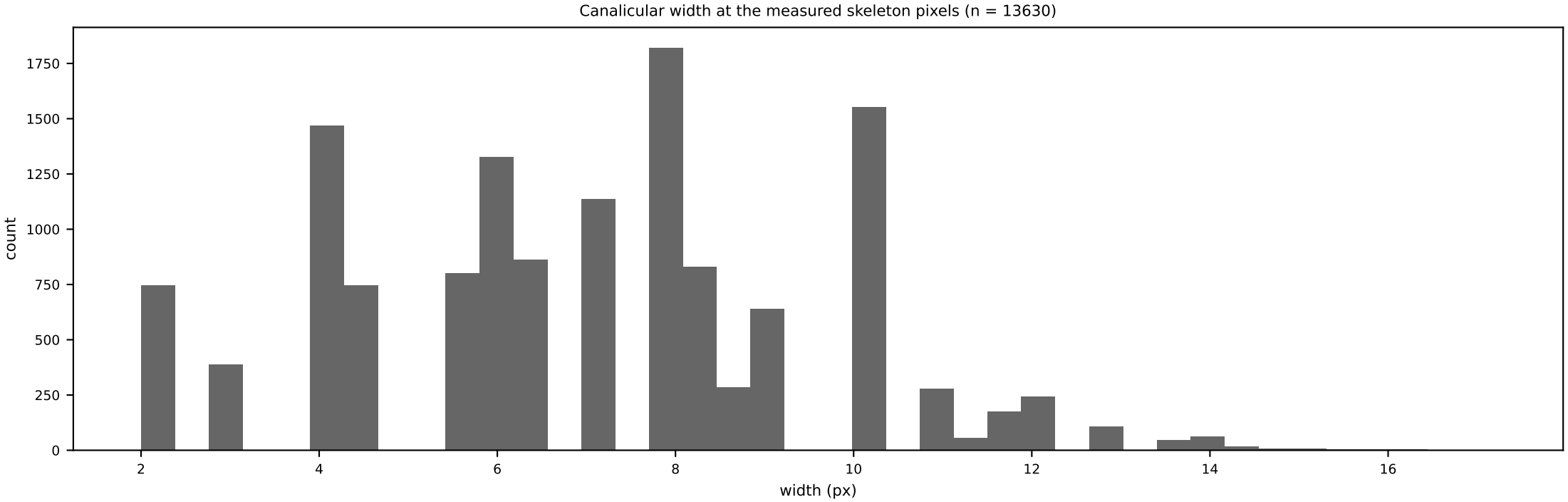
1	1	135	406	17.4853	1	17.4853	6.782
2	12	320	421	401.321	30	13.3774	7.496
3	8	218	510	837.512	41	20.4271	8.6503
4	8	343	755	809.838	50	16.1968	6.2878
5	9	347	687	1878.48	104	18.0623	7.0427
6	11	254	275	296.108	18	16.4504	7.1579
7	9	424	1083	1709.11	89	19.2035	7.0353
8	6	192	446	345.238	17	20.3081	6.6589
9	4	219	405	185.539	16	11.5962	6.8391
10	16	545	972	1116.52	59	18.924	7.842
11	5	141	237	166.255	8	20.7819	7.3126
12	7	348	751	1017.65	49	20.7683	6.0891
13	8	112	254	103.77	7	14.8242	6.8038
14	2	40	40	52.6274	4	13.1569	5.7532
15	5	73	124	46.2132	5	9.2426	6.3638
16	1	26	67	19.4853	1	19.4853	5.3649
17	5	330	688	680.801	31	21.9613	7.6111
18	3	121	254	120.284	6	20.0474	7.602
19	3	104	205	157.368	5	31.4735	5.7226
20	2	35	62	18.4853	1	18.4853	5.146

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	115.083	3546	9848	0.6007	0.566	0.0380711	0.0412266	0.8689
2	169.681	6393	14892	1	1	0.0500548	0.0282702	7.0721
3	185.581	6924	15979	1	1	0.0314847	0.0319169	4.3108
4	418.517	11182	24664	1	1	0.0306743	0.0306114	1.9115
5	534.86	8448	16501	0.7334	0.6731	0.0410748	0.0416338	1.6827
6	728.541	11456	21516	0.6459	0.6063	0.0221718	0.0127812	1.5099
7	221.179	7889	21117	1	1	0.0537457	0.0512857	4.0691
8	140.326	6761	16630	1	1	0.0283982	0.026819	4.2758
9	165.74	5442	13737	1	1	0.0402426	0.0294824	2.4134
10	316.149	10460	25779	1	1	0.0521032	0.0377051	5.0609
11	131.439	3919	7916	1	1	0.0359786	0.0299394	3.8041
12	132.267	6479	16157	1	1	0.053712	0.0464814	5.2923
13	363.747	9538	20842	1	1	0.0117425	0.0121869	2.1993
14	224.528	4228	7038	1	1	0.00946074	0.00568343	0.8908
15	285.22	8488	19536	1	1	0.00860038	0.00634726	1.753
16	652.181	10802	23498	0.7216	0.6773	0.00240696	0.00285131	0.1533
17	676.281	17978	35439	1	1	0.0183558	0.0194136	0.7393
18	308.551	9405	21041	1	1	0.0128655	0.0120717	0.9723
19	247.421	8982	20148	1	1	0.0115787	0.0101747	1.2125
20	163.681	5591	13507	0.7628	0.6662	0.00626006	0.00459021	1.2219

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	66	72	153.054	450.291	3	5	5
2	295	348	355.25	468.576	8	8	6
3	205	428	244.238	573.345	5	7	8
4	216	332	386.433	843.11	9	9	12
5	306	474	386.35	769.671	8	5	8
6	248	269	287.037	308.764	7	7	3
7	353	769	474.262	1215.65	7	11	11
8	184	253	212.297	498.818	3	6	4

9	102	105	233.054	448.363	5	4	7
10	543	767	624.014	1112.56	15	13	12
11	139	164	151.912	260.338	5	4	4
12	298	637	387.25	841.541	6	12	13
13	112	182	120.498	274.238	4	2	5
14	40	40	42.5563	42.5563	1	1	0
15	73	91	82.7696	147.024	1	2	2
16	21	21	29.1421	74.7696	1	0	2
17	190	450	369.049	768.884	6	9	13
18	79	160	129.083	273.853	3	3	6
19	48	71	116.397	225.48	2	2	4
20	24	24	40.4558	70.8406	0	0	1

699_HYP_HP_z20c1-2: distributions, pre-validation, px